Mathematics in CP

Arman Ferdous

Euclidean Algorithm

- Properties of GCD:
- If a > b, GCD(a, b) = GCD(a - b, b)
- Extend this to obtain GCD(a, b) = GCD(b, a % b)

Modular Arithmetic

- $X \sim Y$ if $M \mid (X - Y)$
- $X \sim Y$ if $X - Y = M * k$
- $X \sim Y$ if $X \equiv Y \pmod{M}$
- How do the partitions look like?
- $[0] = \{..., -M, 0, M, 2M, \dots\}$
- $[1] = \{..., 1-M, 1, 1+M, 1+2M, \dots\}$
- …
- $[M - 1] = \{... -1, M-1, 2M-1, \dots\}$
- The values from $[0, M-1]$ are the principal values.
- Principal values can be found by taking the remainder when dividing by M.

Properties of Modular Arithmetic

- https://en.wikipedia.org/wiki/Modular_arithmetic#Basic_properties

Binary Exponentiation

- How to calculate a^b % m?

```
// calculate a^b (mod m)
int powmod (int a, int b, int M) {
    int res = 1;
    while (b) {
        int last_bit = b % 2;
        if (last_bit == 1) {
            res = (res * a) % M;
        }
        b /= 2;
        a = (a * a) % M;
    }
    return res;
}
```

Bézout’s Identity

- https://en.wikipedia.org/wiki/B%C3%A9zout%27s_identity
- For any two integers a and b, there exists positive integers x and y such that

$$ax + by = \gcd(a, b)$$

Extended Euclidean Algorithm

- A way to fish out those x and y values of Bezout.
- Run the normal Euclidean algorithm, but figure out the values of x and y from base cases.
- Suppose you found x_1, and y_1 for GCD(b, a % b).

  Then, $b * x_1 + (a \% b) * y_1 = g$
- Now, we want x and y for:

  $a * x + b * y = g$
- Write a % b = a - floor(a / b) * b and substitute in the first equation.

Linear Diophantine Equations

- Equations of the form:

$$Ax + By = C$$

- How to solve this for integer values only? Geometric representation?
- Has multiple solutions. If (x1, y1) is one solution, then

$$(x1 + k * (B/G), y1 - k * (A/G))$$

  Is also a solution (for integer k).

Modular Multiplicative Inverse

- Now, if we want to find a way to solve $23 \equiv 4x \pmod 8$, by dividing by 4 on both sides, what we really want to find is a number y to multiply on both sides, such that the 4 on the RHS cancels out.
- Modular Multiplicative Inverse: of a number x under modulo M is a number y such that $x * y = 1 \pmod M$
- So, $x * y - 1 = M * k$
- Which gives, $x * y + M * (-k) = 1$
- We know x and M, we solve for y and (-k) using egcd and use y as the inverse!
- So when do inverses exist?

Fermat's Little Theorem

- For any prime number p, if p does not divide a then

$$a^{p-1} \equiv 1 \pmod{p}$$

  How?
- Take S = {1, 2, …, p - 1}.
- Now let aS = {1a, 2a, …, (p-1)a} all modulo p
- Show that aS is just a permutation of S.
- Then, 1 * 2 * … * (p-1) ≡ (1a) * (2a) * … * ((p-1)*a) (mod p)

Why do we need FLT?

- FLT can help reduce large powers into smaller values.
- If we need to calculate a^b % p, where b is too large, we can't simply calculate a^(b%p) % p, this would be incorrect.
- Using FLT, we can raise our modulo towards the exponent!
- Example: evaluate

$$15^{8^{13}} \pmod{11}$$

Euler's Totient Function

- φ(n) = number of integers < n which are coprime with n.
- Simply, φ(n) is the number of elements modulo n, which have inverses.
- How to calculate φ(n)?
- Let

$$n = p_1^{e_1} \cdots p_k^{e_k}$$

Then

$$\varphi(n) = n \times \prod_{i=1}^{k} (1 - \frac{1}{p_i})$$

Euler's Theorem

- If a, n are coprime (i.e. gcd(a, n) = 1), then

$$a^{\varphi(n)} \equiv 1 \pmod{n}$$

Chinese Remainder Theorem (simplified)

- Let n and m be integers such that gcd(n, m) = 1.
- Then, every integer x in [0, nm - 1] gets mapped to a unique pair of integers

$$(x \bmod n, x \bmod m)$$

- In reverse, this implies that if you are given

$$x \equiv y1 \pmod{n} \quad \text{and } x \equiv y2 \pmod{m}$$

  Then there exists an unique solution x (mod nm)
- How to solve? CRT wiki

Using CRT to prove that phi is multiplicative

- What is a multiplicative function?
- If a function f has the following properties:
  f(1) = 1, and
  If n, m are co-prime then f(nm) = f(n) * f(m)
  Then f is multiplicative.
- Do we know any multiplicative functions?

Using CRT to prove that phi is multiplicative

- Now, gcd(x, mn) = 1 implies that gcd(x, m) = 1 AND gcd(x, n) = 1.
- And, gcd(x, m) = gcd(x mod m, m)
  gcd(x, n) = gcd(x mod n, n)
- Hence, φ(mn) = φ(m) * φ(n) as CRT guarantees that every x in [0, mn - 1] has unique (x mod m, x mod n) pairs.
- Now how to calculate φ(p^k)?

Other useful properties of Euler's Totient Function

- Divisor sum property:

$$\sum_{d|n} \varphi(d) = n$$

- Symmetry property:

$$\sum_{\gcd(i,n)=1, 1 \le i \le n} i = \frac{n \times \varphi(n)}{2}$$

Other useful properties of Euler's Totient Function

- Euler's Theorem:

$$\gcd(a, m) = 1 \implies a^b \equiv a^{b \mod \varphi(m)} \pmod{m}$$

- Generalized Euler's theorem, (when b > phi(m))

$$a^b \equiv a^{(b \mod \varphi(m)) + \varphi(m)} \pmod{m}$$

Other useful properties of Euler's Totient Function

- $\varphi(n)$ is even for any $n > 2$. How?
- $\varphi(n)$, $\varphi(\varphi(n))$, $\varphi(\varphi(\varphi(n)))$ … reaches 1 in within $O(\lg n)$ steps.

GCD sum and LCM sum problems

$$\sum_{i=1}^{n} \gcd(i,n) = \sum_{d|n} d \cdot \varphi(\frac{n}{d})$$

$$\sum_{i=1}^{n} \operatorname{lcm}(i,n) = \frac{n}{2}[1 + \sum_{d|n} d \cdot \varphi(d)]$$

Calculating nCr % p

- https://cses.fi/problemset/task/1079

https://codeforces.com/gym/106178/problem/F

https://cses.fi/problemset/task/2185

https://codeforces.com/contest/1916/problem/B

https://codeforces.com/contest/2226/problem/B